

Linux Terminal

Quick Reference Tutorial

Navigating the Filesystem

```
pwd # Print current directory
ls # List files in current directory
ls -la # List all files (including hidden) with details
cd /home/user # Change to an absolute path
cd Documents # Change to a relative path
cd .. # Go up one directory
cd ~ # Go to your home directory
cd - # Go back to previous directory
```

Managing Files & Folders

```
mkdir projects # Create a directory
mkdir -p a/b/c # Create nested directories
touch file.txt # Create an empty file
cp file.txt backup.txt # Copy a file
cp -r dir1 dir2 # Copy a directory recursively
mv file.txt /tmp/ # Move a file (also used to rename)
rm file.txt # Delete a file
rm -rf mydir # Delete a directory and contents
```

Viewing & Editing Files

```
cat file.txt # Print file contents
less file.txt # Scroll through a file (q to quit)
head -n 20 file.txt # Show first 20 lines
tail -f log.txt # Follow a file in real time (great for logs)
nano file.txt # Simple terminal text editor
```

Searching

```
find . -name "*.txt" # Find files by name
grep "error" log.txt # Search inside a file
grep -r "TODO" ./src # Search recursively in a directory
```

Permissions

```
chmod +x script.sh # Make a file executable
chmod 755 file # Set permissions (owner: rwx, others: r-x)
chown user:group file # Change file owner
sudo command # Run a command as superuser
```

Processes

```
ps aux # List all running processes
top / htop # Live process monitor (q to quit)
kill 1234 # Kill process by PID
kill -9 1234 # Force kill
```

Note: Ctrl+C to interrupt a running command | Ctrl+Z to suspend | bg to resume in background

SSH — Remote Access

```
ssh user@hostname # Connect to a remote server
ssh user@192.168.1.10 # Connect by IP
ssh -p 2222 user@hostname # Custom port

scp file.txt user@host:/path/ # Upload a file
scp user@host:/path/file.txt .
# Download a file

scp -r dir/ user@host:/path/ # Copy a whole directory

ssh-keygen # Generate a key pair
ssh-copy-id user@host # Copy public key to server
```

Handy Shortcuts

Shortcut	Action
Tab	Autocomplete
↑ / ↓	Navigate command history
Ctrl+R	Search command history
Ctrl+L	Clear terminal
!!	Repeat last command
command &	Run command in background

Piping & Redirection

```
ls -la | grep ".txt" # Pipe output into another command
cat file.txt | sort # Sort lines
command > output.txt # Write output to file (overwrite)
command >> output.txt # Append output to file
command 2> errors.txt # Redirect errors to file
```

Pro tip: `man <command>` shows the full manual for any command (e.g. `man ssh`). Press `q` to exit.